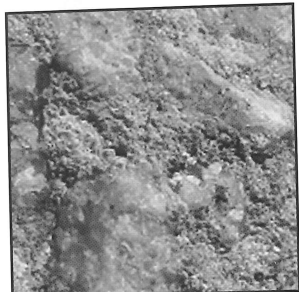


Experiment 31: Acid reactions with some metal compounds



Concrete

Notes

Acids are substances that donate protons, and in aqueous solutions produce hydronium ions, H_3O^+ .

Bases are substances that accept protons, and in aqueous solutions produce hydroxide ions, OH^- .

Understanding the properties of acids and bases is vital to the understanding of the effect of water that has a low pH on metal compounds such as metal oxides, hydroxides and carbonates found in rocks and soils as well as in many modern structures making use of concrete and mortar.

Commonly available acids include spirits of salts (hydrochloric acid), battery acid (sulfuric acid) and vinegar (acetic acid), and commonly available bases include soda ash (sodium carbonate), baking soda (sodium hydrogencarbonate), caustic soda (sodium hydroxide) and ammonia.

In this experiment you will investigate the effect of acids (using hydrochloric acid and vinegar) on some oxides, hydroxides, carbonates and hydrogen carbonates commonly found in rocks, soils and building and household materials.

Equipment

Bunsen burner

dropper

test tubes (7)

matches

hydrochloric acid, HCl , 2 mol L^{-1} (40 mL)

vinegar (10 mL)

limewater, $Ca(OH)_2$ (40 mL)

about 1 g each of the metal oxides and hydroxides:

copper(II) oxide $[CuO]$ found as the mineral cuprite, a copper ore

iron(III) oxide $[Fe_2O_3]$ found as the mineral hematite, an iron ore and in rust

zinc oxide $[ZnO]$ found as the mineral zincite, a zinc ore

lead(II) oxide $[PbO]$ found in the mineral litharge, a lead ore

aluminium hydroxide $[Al(OH)_3]$ found in the mineral bauxite, an aluminium ore

calcium hydroxide $[Ca(OH)_2]$ found in concrete and mortar.

about 1 g each of the metal carbonates and hydrogen carbonates:

calcium carbonate $[CaCO_3]$ marble chips, limestone used as a building material

sodium carbonate $[Na_2CO_3]$ soda ash, used in washing powder, as a swimming pool buffer and in the making of glass

sodium hydrogencarbonate $[NaHCO_3]$ baking soda used as a leavening agent in baking.

about 10 mL of sodium hydroxide (caustic soda) $[NaOH]$ 2 mol L^{-1} , a strong base used in the refining of bauxite to alumina

blue litmus paper

red litmus paper

litmus solution

Procedure

1. Draw up a table that can be used to summarise the observations for the experiments you will perform. Your table should include the following headings:
 - (a) the reactants involved in the reaction you are performing;
 - (b) your observations of the reaction: these observations should be thorough and clearly written;
 - (c) products identified from your observations and an explanation of how your observations allowed you to identify those products.
2. Carry out the following reactions carefully. All members of a group should make observations of each reaction, then write these into their own prepared table.

SAFETY NOTE:

Acids and bases can react with a large variety of materials including your skin, clothing and books. Take care not to spill acids and bases, particularly onto your skin. If spills do occur wash the affected area with large quantities of water. You must wear safety glasses at all times during this experiment.

3. Solutions of hydrochloric acid and sodium hydroxide are both colourless. Make sure that containers used to store these are labelled. You can check whether a solution is an acid or a base by using an indicator such as litmus indicator paper. To establish the test, do the following and record your observations.
 - (a) Place a drop of $2 \text{ mol L}^{-1} \text{ HCl}$ onto pieces of red and blue litmus paper.
 - (b) Repeat using $2 \text{ mol L}^{-1} \text{ NaOH}$.

A Neutralisation of a strong acid with a strong base

1. Place 10 drops of $2 \text{ mol L}^{-1} \text{ HCl}$ into a test tube and add three drops of litmus solution or small pieces of red and blue litmus paper. Note the colour of the solution.
2. Add $2 \text{ mol L}^{-1} \text{ NaOH}$ dropwise until there is a permanent colour change. Record observations.

B Reaction of acids with metal oxides and hydroxides

1. Place a small quantity (about the size of two match heads) of CuO in a test tube. Add 4 to 5 mL of $2 \text{ mol L}^{-1} \text{ HCl}$, gently heat the mixture but do not boil it, then allow it to stand.
2. Record your observations.
3. Repeat step 1 and 2 with all the other oxides and hydroxides.

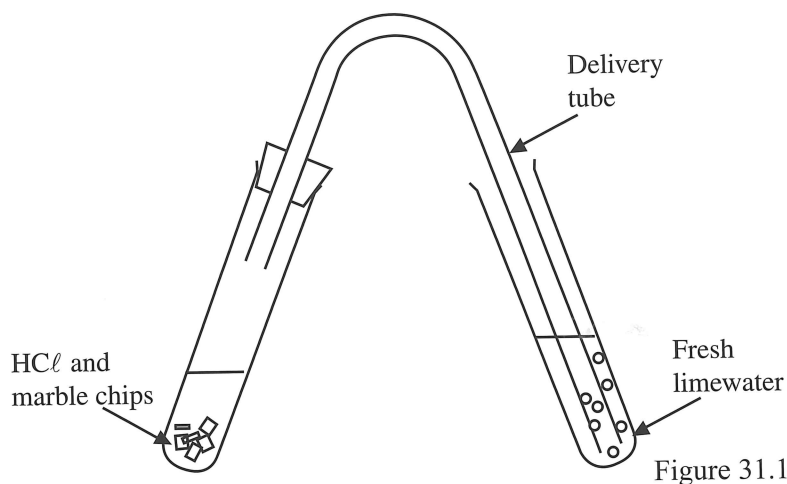
Notes

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Notes

C Reaction of acids with metal carbonates and hydrogencarbonates

1. Place a small quantity (about the size of two match heads) of CaCO_3 in a test tube. Add 2 to 3 mL of 2 mol L^{-1} HCl . Set up the test tube as shown in figure 31.1.



2. Record your observations.
3. Repeat step 1 and 2 with Na_2CO_3 and NaHCO_3 .
4. Repeat steps 1 and 2 with CaCO_3 using vinegar in place of the HCl solution.

Processing the results and questions

1. Write an ionic equation for each of the reactions you observed.
2. Compare and contrast the reactions of sodium hydroxide and ammonia with hydrochloric acid.
3. Why do you think it was advisable to warm some of the solutions?
4. Compare and contrast the reactions of vinegar and hydrochloric acid with calcium carbonate.
5. Acid rain is a problem in industrial areas that burn sulfur-containing coal like Europe, China and Russia. Research the pH of acid rain and the effect of acid rain on limestone, marble and sandstone buildings. Use one or more of the reactions observed in this experiment to either support or refute your research findings regarding the impact of acid rain on buildings.